

Register No.	
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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION ENGINEERING

FT –I

Academic Year: 2025-26

Program offered: M.Tech (AR & E&CE)

Year / Sem: I/II

Course Title: Embedded control systems

Max. Marks: 40

Course Code: 21EIE602T

Duration: 90 Min.

Date of Exam: 19/2/2026

PART A (Any 2) (2X20=40 marks)

Sl. No	Question
1	a) Explain rate monotonic scheduling (RMS) with its assumptions. (5 Marks) b) Explain earliest deadline first (EDF) scheduling and its advantages. (5 Marks) c) Compare preemptive and non-preemptive scheduling in real-time systems. (5 Marks) d) Define task, process, and thread in the context of real-time systems. (5 Marks)
2	a) Explain peripheral interfacing with suitable examples. (5 Marks) b) Elucidate SPI, I2C & UART protocol and its applications. (15 Marks)
3	a) Explain TCP/IP stack with reference to embedded systems. (10 Marks) b) Discuss the role of bus structure. (10 Marks)